BEE 4750 Lab 2: Monte Carlo

Name:

ID:

> **Due Date**
>
> Thursday, 10/02/25, 9:00pm

Setup

The following code should go at the top of most Julia scripts; it will load the local package environment and install any needed packages. You will see this often and shouldn't need to touch it.

```julia
import Pkg
Pkg.activate(".")
Pkg.instantiate()
```

```
  Activating project at `~/4750/lab-2-d8rha`
```

```julia
using Random # random number generation
using Distributions # probability distributions and interface
using Statistics # basic statistical functions, including mean
using Plots # plotting
```

Overview

In this lab, we will conduct a Monte Carlo experiment and explore how sensitive Monte Carlo estimates are to the underlying probability distribution(s).

You should always start any computing with random numbers by setting a "seed," which controls the sequence of numbers which are generated (since these are not *really* random, just "pseudorandom"). In Julia, we do this with the `Random.seed!()` function.

```
Random.seed!(1)
```

```
TaskLocalRNG()
```

It doesn't matter what seed you set, though different seeds might result in slightly different values. But setting a seed means every time your notebook is run, the answer will be the same.

> **Seeds and Reproducing Solutions**
>
> If you don't re-run your code in the same order or if you re-run the same cell repeatedly, you will not get the same solution. If you're working on a specific problem, you might want to re-use `Random.seed()` near any block of code you want to re-evaluate repeatedly.

Probability Distributions and Julia

Julia provides a common interface for probability distributions with the `Distributions.jl` package. The basic workflow for sampling from a distribution is:

1. Set up the distribution. The specific syntax depends on the distribution and what parameters are required, but the general call is the similar. For a normal distribution or a uniform distribution, the syntax is

   ```
   # you don't have to name this "normal_distribution"
   #   is the mean and   is the standard deviation
   normal_distribution = Normal( ,  )
   # a is the upper bound and b is the lower bound; these can be set to +Inf or -Inf for ar
   uniform_distribution = Uniform(a, b)
   ```

   There are lots of both univariate and multivariate distributions, as well as the ability to create your own, but we won't do anything too exotic here.

2. Draw samples. This uses the `rand()` command (which, when used without a distribution, just samples uniformly from the interval $[0, 1]$.) For example, to sample from our normal distribution above:

   ```
   # draw n samples
   rand(normal_distribution, n)
   ```

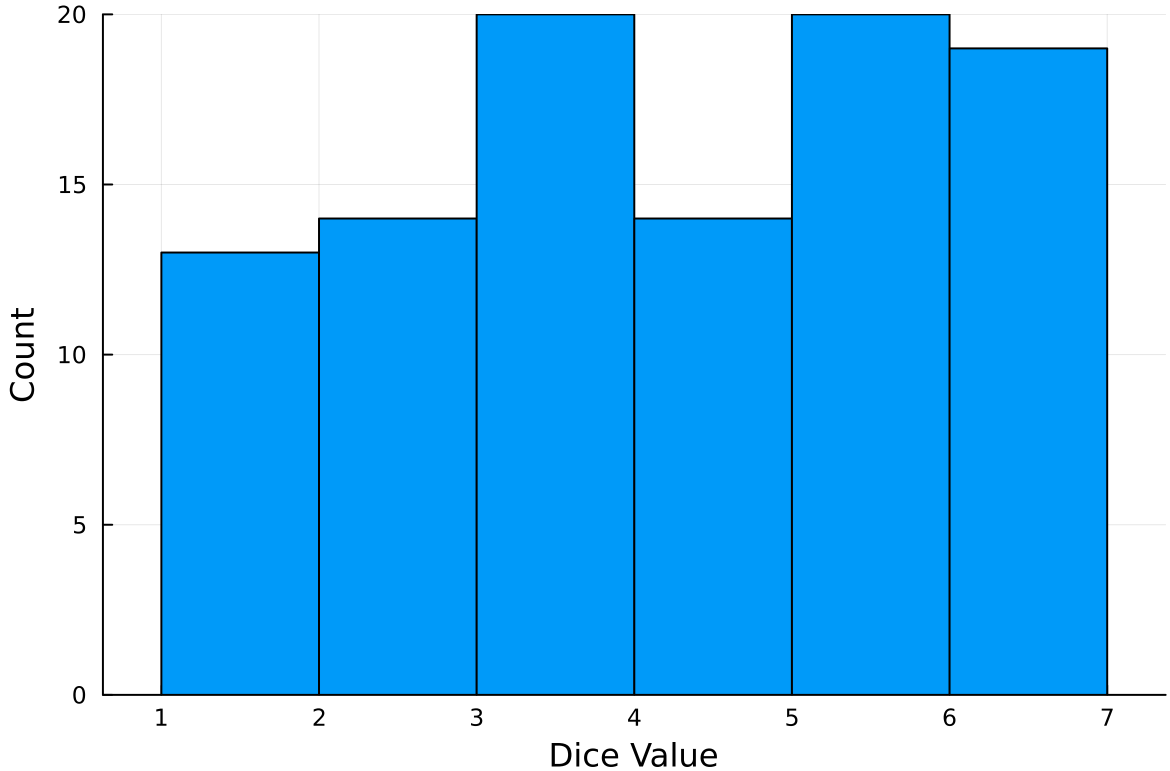
Putting this together, let's say that we wanted to simulate 100 six-sided dice rolls. We could use a Discrete Uniform distribution.

```
dice_dist = DiscreteUniform(1, 6) # can generate any integer between 1 and 6
dice_rolls = rand(dice_dist, 100) # simulate rolls
```

```
100-element Vector{Int64}:
 1
 3
 5
 4
 6
 2
 5
 5
 5
 2

 3
 6
 5
 5
 6
 3
 6
 6
 6
```

And then we can plot a histogram of these rolls:

```
histogram(dice_rolls, legend=:false, bins=6)
ylabel!("Count")
xlabel!("Dice Value")
```



Instructions

Remember to:

- Evaluate all of your code cells, in order (using a `Run All` command). This will make sure all output is visible and that the code cells were evaluated in the correct order.
- Tag each of the problems when you submit to Gradescope; a 10% penalty will be deducted if this is not done.

Problem

A common engineering problem is to quantify flood risk, which is typically computed by propagating a flood hazard distribution through a *depth-damage function* relating flood depths to economic damages. A reasonable depth-damage function for a house without a basement is a bounded logistic function,

$$d(h) = \mathbb{1}_{h>0} \frac{L}{1 + \exp(-k(h - h_0))},$$

where d is the damage as a percent of total structure value, h is the water depth (relative to the house's ground floor elevation) in m, $\mathbb{1}_{x>0}$ is the indicator function, L is the maximum loss in USD, k is the slope of the depth-damage relationship, and h_0 is the inflection point. We'll assume $L = \$200,000$, $k = 0.8$, and $h_0 = 3$.

For this problem, suppose that we have two different probability distributions characterizing annual maxima flood depths:

1. $h \sim LogNormal(1.2, 0.3)$;
2. $h \sim GeneralizedExtremeValue(3, 0.9, -0.15)$.

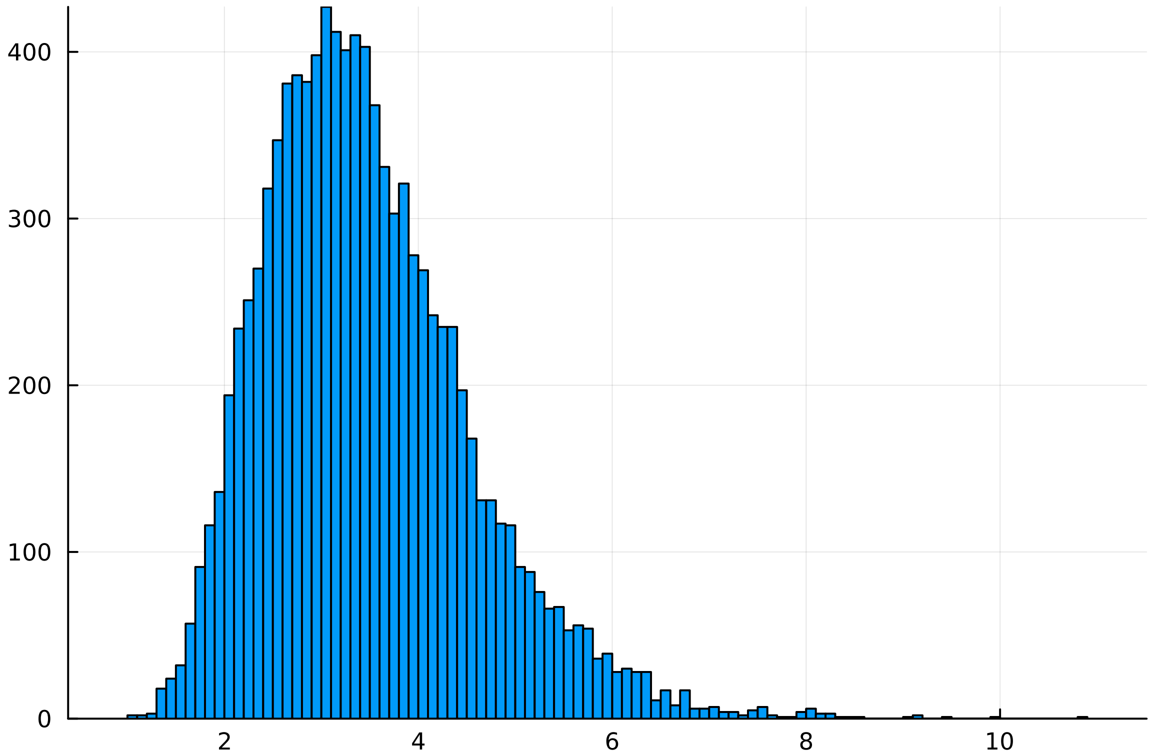
Plot histograms of samples from these distributions and conduct a Monte Carlo experiment for annual maximum flood damages using each. Answer the following questions:

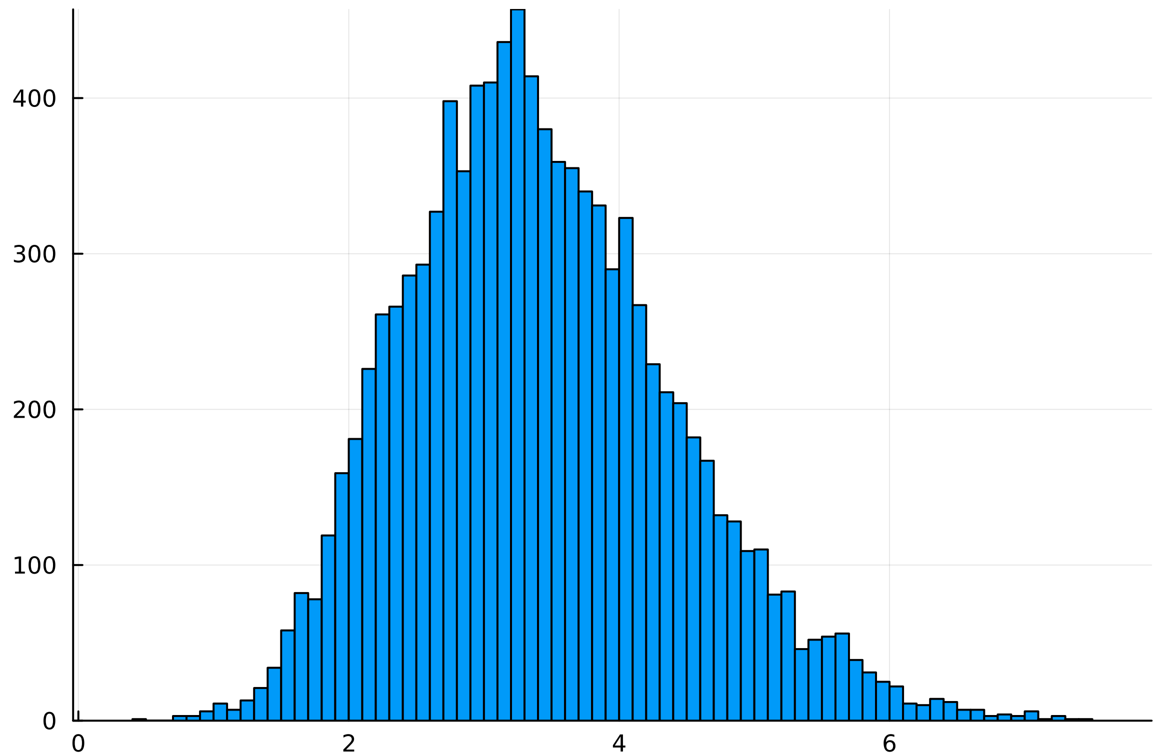
- What are the Monte Carlo estimates of the expected value and the 99% quantile for the annual maximum damage the structure would suffer for each of these flood hazard distributions?
- How did you decide on your sample size?
- Why do you think the estimates differed or did not differ (you can also plot the depth-damage function to compare with the samples)?
- How might you decide (based on getting additional information, if needed) which distribution is "better"?

```
function d(h);
    L = 200000
    k = 0.8
    h0 = 3.0
    if h > 0
        L/(1 + exp(-k * (h - h0)))
    else
        return 0
    end
end

log_normal = LogNormal(1.2,0.3)
sample_log_normal = rand(log_normal, 10000)

damage_distribution_log_norm = zeros(10000)

for k in eachindex(sample_log_normal)
    damage_distribution_log_norm[k] = d(sample_log_normal[k])
end

# print(damage_distribution_log_norm)
```

```
gen_ext_val = GeneralizedExtremeValue(3,0.9,-0.15)
sample_gen_ext_val = rand(gen_ext_val, 10000)

damage_distribution_gen_ext_val = zeros(10000)

for k in eachindex(sample_gen_ext_val)
    damage_distribution_gen_ext_val[k] = d(sample_gen_ext_val[k])
end

# print(damage_distribution_gen_ext_val)

# --- Monte Carlo estimates ---
mean_log = mean(damage_distribution_log_norm)
q99_log  = quantile(damage_distribution_log_norm, 0.99)

mean_gev = mean(damage_distribution_gen_ext_val)
q99_gev  = quantile(damage_distribution_gen_ext_val, 0.99)

display("Mean (Log Normal): $(mean_log)")
display("99th Quantile (Log Normal): $(q99_log)")
display("Mean (Generalized Extreme Value): $(mean_gev)")
display("99th Quantile (Generalized Extreme Value): $(q99_gev)")

p = histogram(sample_log_normal; bins=100, legend=false)
# xlabel!(p, "Annual Maximum Flood Damages using Log Normal Distribution")
# ylabel!(p, "Water Depth")
display(p)

p = histogram(sample_gen_ext_val; bins=100, legend=false)
# xlabel!(p, "Annual Maximum Flood Damages using Generalized Extreme Value Distribution")
# ylabel!(p, "Water Depth")
display(p)
```

```
"Mean (Log Normal): 115112.85644248554"

"99th Quantile (Log Normal): 189104.17774290868"

"Mean (Generalized Extreme Value): 114014.94240286636"

"99th Quantile (Generalized Extreme Value): 183489.8890106969"
```


400
300
200
100
0
2
4
6
8
10



- What are the Monte Carlo estimates of the expected value and the 99% quantile for the annual maximum damage the structure would suffer for each of these flood hazard distributions?

Mean (Log Normal): 115112.85644248554

99th Quantile (Log Normal): 189104.17774290868

Mean (Generalized Extreme Value): 114014.94240286636

99th Quantile (Generalized Extreme Value): 183489.8890106969

- How did you decide on your sample size?

I chose a sample size that best resembled a noticeable logarithmic normal distribution without being too computationally strenuous.

- Why do you think the estimates differed or did not differ (you can also plot the depth-damage function to compare with the samples)?

I think the difference in the estimates could be attributed to the mathematical properties of each of the two methods, resulting in different distribution shapes. However, the estimates did not differ by much since the sample size was large enough that the distributions are similar to normal.

- How might you decide (based on getting additional information, if needed) which distribution is "better"?

To decide which distribution is better, a data fit could be used to determine which distribution is closer to what the real results should display. This could also be through theoretical analysis to determine which mathematical processes within the two distributions would best reflect real conditions and such.

References

Put any consulted sources here, including classmates you worked with/who helped you.

Consulted Sources

Since I was having trouble with identifying the correct syntax for Julia in the for loop portion of my code, I used ChatGPT to debug my code.

I inputted

"I'm trying to use a for loop to replace each index in the damage_distribution_log_norm vector with the randomly generated value from sample_gen_ext_val inputted into the d(h) function and replace that value into the index instead. Can you help me solve this in Julia?"

and it outputted

"damage_distribution = zeros(100) for i in eachindex(sample_gen_ext_val) damage_distribution[i] = d(sample_gen_ext_val[i]) end println(damage_distribution)"

I was able to then identify that I had to use eachindex() instead of range(length()) which I previously tried to use in the for loop.

I also worked on this lab with Evan Sims (ems452) during class, and he helped me confirm some of the logic of how I set up the for loops.